

YOSHEE JAIN

Undergraduate Research Assistant
 Siebel School of Computing and Data Science
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EDUCATION

University of Illinois Urbana-Champaign

Champaign-Urbana, IL

- Bachelor of Science in Computer Science
- Minor in Statistics
- Completed Data Science Certification

May 2026**GPA: 4.0/4.0**

- **Thesis:** Towards Identifying Domain-Specific Programming Plans at Scale: Needs, Challenges, and Solutions

RESEARCH OBJECTIVE

My research interests lie in designing interactive learning environments that help people from diverse backgrounds learn and use computing. I understand how students anticipate using computational knowledge in their future careers using HCI methods. I then leverage learning science research to theorize and prototype pedagogical scaffolds that meaningfully support learners' goals. Finally, using AI/ML techniques, I develop and evaluate human-centered systems that implement these instructional interventions at scale.

Topics: Human-Centered AI, Human-Computer Interaction, CS Education, Human-AI Collaboration, Cognitive Science

PUBLICATIONS

Under Review

* Indicates Equal Contribution.

- [J.3] Linguistic Comparison of AI- and Human-Written Responses to Online Mental Health Queries.

Nature: Artificial Intelligence. Submitted April 2025.

K. Saha, Y. Jain, M. De Choudhury.

Heavily Peer-Reviewed Conference Proceedings

- [C.3] Uncovering the Internet's Hidden Values: An Empirical Study of Desirable Behavior Using Highly-Upvoted Content on Reddit.
 2026 International AAAI Conference on Web and Social Media (ICWSM 2026)

A. Goyal, C. Lambert, Y. Jain, E. Chandrasekharan.

- [C.2] PLAID: Supporting Computing Instructors to Identify Domain-Specific Programming Plans at Scale.
 2025 ACM CHI Conference on Human Factors in Computing Systems (CHI 2025)

Y. Jain, M. Demirtas*, K. Cunningham.*

- [C.1] Exploring the Potential of Large Language Models for Estimating the Reading Comprehension Question Difficulty.
 2025 International Conference on Human-Computer Interaction (HCII 2025)

Y. Jain, John Hollander, Amber He, Sunny Tang, Liang Zhang, John Sabatini.

Journal Articles

- [J.2] Online Communities as a Support System for Alzheimer's and Dementia Care: Large-scale Exploratory Study.
 JMIR Aging 2025

S. Kaliappan, C. Liu, Y. Jain, R. Karkar, K. Saha.

- [J.1] AI vs Humans for Online Support: Comparing the Language of Responses from LLMs and Online Communities of Alzheimer's Disease.

ACM Transactions on Computing for Healthcare 2025

K. Saha, Y. Jain, C. Liu, S. Kaliappan, R. Karkar.

Posters

- [P.3] Examining the Efficacy of Hashtag-Based Bans on Instagram.
2025 ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2025)
E. Mok, Y. Jain*, S. Gottiparthi*, E. Chandrasekharan.*
- [P.2] Investigating How Gilds Were Employed On Reddit.
2024 ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2024)
C. Lambert, Y. Jain, K. Saha, E. Chandrasekharan.
- [P.1] Towards Methods for Identifying High-Quality Domain-Specific Programming Patterns.
2023 ACM Conference on International Computing Education Research V.2 (ICER 2023)
Y. Jain, K. Cunningham.

Symposiums & Lightly-Peer Reviewed Conferences

- [S.2] Exploring the Use of LLMs to Generate Contextualized Worked Examples From Programming Patterns.
TAPIA Conference 2025
Y. Jain, K. Cunningham.
- [S.1] PLAID: Supporting Computing Instructors to Identify Domain-Specific Programming Plans at Scale.
Symposium on AI, Education, and the Learning Sciences, Northwestern University.
Y. Jain, M. Demirtas, K. Cunningham.

Bachelor's Thesis

- [T.1] Towards Identifying Domain-Specific Programming Plans at Scale: Needs, Challenges, and Solutions
Bachelor's Thesis.
Y. Jain

RESEARCH EXPERIENCE

- Undergraduate Summer Research Intern**

Jun 2025 – Present

Korea Advanced Institute of Science and Technology

Dr. Juho Kim, Dr. Seungju Kim, Saelyne Yang, Chen Zhou

 - Designing a personalized video learning interface that simplifies video content (video speed, density of visuals, complexity of information, audio clarity) in real-time based on the individual cognitive needs of adults with intellectual disabilities.
 - Area of Work: Human-Computer Interaction, Computer Vision, Multimodal LLMs, Education
- Undergraduate Summer Research Assistant**

May 2025 – Present

Federal Institute of Technology Zurich (ETH Zürich)

Dr. April Yi Wang, Dr. Zihan Wu, Dr. Heejin Do

 - Developing a tool that integrates students' explanations of algorithms with input-output traces of examples, to reduce LLM misinterpretations of these natural language descriptions and enhance the reliability of autograding.
 - Area of Work: Human-Computer Interaction, AI, CS Education, Experimental Research, Mixed-Methods Research
- Undergraduate Research Assistant**

Jan 2023 – Present

University of Illinois Urbana-Champaign

Dr. Katie Cunningham

 - Building a learning tool to assist CS majors in explaining technical concepts to non-technical audiences, and non-CS majors in talking about technical ideas, to improve communication related to computing for workplace interactions.
 - Area of Work: CS Education, Human-Computer Interaction, Agentic AI, Experimental Research, Mixed-Methods Research
 - Conducted a formative study to uncover challenges faced by instructors when redesigning courses using pattern-based pedagogy and developed and evaluated a system using human-AI collaboration to support pattern editing. **[C.2, P.1, S.1, T.1]**
 - Area of Work: CS Education, Human-Computer Interaction, AI, Experimental Research, Mixed-Methods Research
 - Assessed the capabilities of large language models in generating contextualized worked examples from programming patterns to design curricula for learners from diverse backgrounds. **[S.2]**
 - Area of Work: CS Education, Human-Computer Interaction, AI, Experimental Research, Mixed-Methods Research
- Undergraduate Research Assistant**

Aug 2023 – Aug 2025

University of Illinois Urbana-Champaign

Dr. Eshwar Chandrasekharan, Charlotte Lambert

- Evaluated Instagram’s moderation approach and built a classifier that outperformed it by reducing wrongful penalization. [P.3]
- Area of Work: Human-Computer Interaction, Multimodal LLMs, Social Computing, Data Science, Machine Learning
- Designed a quasi-experimental study on Reddit to assess how positive feedback on content (e.g., upvotes on post replies) affects users’ longitudinal behavior. [P.2]
- Area of Work: Human-Computer Interaction, Social Computing, Data Science, Causal Inference
- **Undergraduate Research Assistant** Aug 2023 – May 2025
University of Illinois Urbana-Champaign *Dr. Koustuv Saha, Dr. Ravi Karkar*
 - Compared LLM responses to queries from Alzheimer’s support forums with human-authored replies; integrated inductive coding and large-scale psycholinguistic and lexico-semantic analyses. [J.2, J.1]
 - Area of Work: Human-Computer Interaction, Social Computing, Data Science, AI, Mixed-Methods Research
- **Undergraduate Research Intern** May 2024 – Aug 2024
Carnegie Mellon University *Dr. Jionghao Lin, Dr. John Sabatini, Dr. John Hollander, Dr. Liang Zhang*
 - Benchmarked LLMs’ ability to estimate reading comprehension-related question difficulty against human annotations. [C.1]
 - Area of Work: Human-Computer Interaction, AI, Education, Machine Learning
 - Created an intelligent tutoring system (ITS) using a multi-agent, RAG-based, tutor-learner simulation that adjusts question difficulty in response to learner performance.
 - Area of Work: Human-Computer Interaction, AI, Education

HONORS, AWARDS, FELLOWSHIPS, AND SCHOLARSHIPS

- **Finalist in the ACM Undergraduate Student Research Competition** 2025
One of the top five finalists of the ACM Undergraduate Student Research competition at TAPIA 2025.
- **Outstanding Course Assistant Award** 2025
One of the five recipients of the Outstanding Course Assistant awards for Spring 2025, awarded for excellent instruction in CS 102.
- **Jeffrey P. Blahut Memorial Scholarship** 2025
One of two recipients of the Jeffrey P. Blahut Memorial Scholarship (\$4,000), awarded for academic excellence in Computer Science.
- **CHP Outstanding Senior Award** 2025
Selected to receive the award (\$1,000) in recognition of my academic record and active participation in the Campus Honors Program.
- **John and Dorothy Durkin Women in Engineering Scholarship** 2025
Recipient of the scholarship for \$5,000 (2025-26) that supports women contributing to the advancement of engineering.
- **Harold and Ruth Hayward/Tau Beta Pi Award Fund** 2025
Recipient of the Harold and Ruth Hayward Scholarship (\$3,000), awarded for excellence, leadership, and service in engineering.
- **Illinois Engineering Achievement Scholarship** 2025
Recipient of the scholarship for \$4,000 (2025-26), \$8,000 (2024-25) and \$1,000 (2023-24) for demonstrated academic excellence.
- **Illinois Engineering Outstanding Scholarship** 2025
Honored with the scholarship, \$1,000 for 2025 and \$2,000 annually 2024–25, for exemplary academic performance in engineering.
- **Conference Travel Grants** 2025
Awarded the Conference Travel Grants from OUR (\$400) and CHP (\$400) to support presentation of research at CHI 2025 in Japan. Also awarded \$400 from CHP for CSCW 2025.
- **Dean’s List** 2025
Dean’s List honoree for six consecutive semesters (FA22–SP25), awarded to the top 20% of students in the College of Engineering.
- **Summer Research Award** 2024
Awarded the CHP Summer Research Fellowship (\$2,000) for independent, faculty-guided research at UIUC.
- **Distinguished Undergraduate Researcher Certificate** 2024
Earned the certificate for exceptional commitment to research, communication, and scholarly contribution at UIUC.

- Distinguished Active Member, Tau Beta Pi (The Engineering Honor Society)**
2024
Recognized as a Distinguished Active Member for sustained engagement, leadership, and service to the Illinois Alpha chapter.
- Chancellor’s Scholar**
2023
Admitted to the Campus Honors Program (~550 students) for academic excellence and interdisciplinary engagement.
- Edmund J. James Scholar**
2023
Designated James Scholar in the Grainger College of Engineering for academic excellence, curricular engagement, and leadership.

TEACHING EXPERIENCE

- Course Assistant**
Fall 2025
CS 101: Intro Computing: Engrg & Sci.
University of Illinois Urbana-Champaign
 - Assisted Prof. Max Fowler and Prof. Nico Ritschel in organizing Python-based labs for non-CS engineering majors, supporting applied computational skill development.
 - Conducted comprehensive review sessions to strengthen conceptual understanding and exam preparedness.
 - Engineered randomized question delivery in PrairieLearn, enhancing grading equity and effectiveness.
- Course Assistant**
Spring 2025
CS/INFO 102: Little Bits to Big Ideas
University of Illinois Urbana-Champaign
 - Aided Prof. Katie Cunningham and Prof. Ryan Cunningham with lectures, labs, and grading in an application-based computing course for non CS majors from diverse majors, including Advertising, Biology, Law, and more.
 - Led exam review sessions to help students from diverse majors learn applied computing, like web development and scraping.
 - Helped develop contextualized lab problems to reinforce the relevance and real-world value of computing concepts.
- Project Manager**
Spring 2023
CS 124 Honors: Introduction to Computer Science I
University of Illinois Urbana-Champaign
 - Mentored three biology students in developing a machine learning backend with a web interface in Android Studio.
 - Advised on technical and logistical challenges throughout project planning and development phases.

SERVICE

- CS Student Ambassadors/Research Scholars**
Jan 2023 - Present
University of Illinois Urbana-Champaign
 - Engaging in the CS department’s outreach, engagement, and recruiting efforts for empowering women in computer science, and thus broadening participation in computing; participated in research fairs and visit day events, organized award ceremonies.
- Content Team Co-Director**
Feb. 2023 - Oct 2024
Reflections Projections 2024. Association of Computing Machinery.
 - Co-hosted a fireside chat with Mr. Thomas Siebel with 150+ participants in 2024.
 - Organized 10+ tech talks by inviting speakers including Dr. Mark Guzdial from the University of Michigan and Dr. Josh Starmer from StatQuest in 2023.

SKILLS

Languages: Python, C/C++, Java, $\LaTeX$, R, HTML/CSS, JavaScript, Flask, Django, TypeScript, React
 * Taken at UIUC.

Frameworks: Cursor, Visual Studio Code, Git/GitHub, Android Studio, PyCharm, REST APIs

Research Methods: System Design, Qualitative Analysis, Quantitative Research, Mixed Methods Analysis, User Interviews, Quasi-Experimental Studies, and Causal Inference Analysis

Relevant Courses*: Methodological Pluralism (Insight into HCI Methodologies), Computer Science Education Research, Computational Social Science, Statistics and Probability, Statistical Programming (R), Database Systems, Text Mining, User Interface Design, Advanced Data Analysis, Algorithms and Models of Computation, Applied Machine Learning